



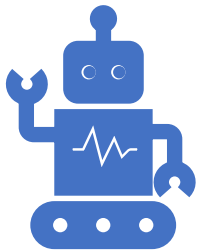
**JASHORE UNIVERSITY OF  
SCIENCE AND TECHNOLOGY**

DEPARTMENT OF

**COMPUTER SCIENCE AND ENGINEERING**

CSE 3201

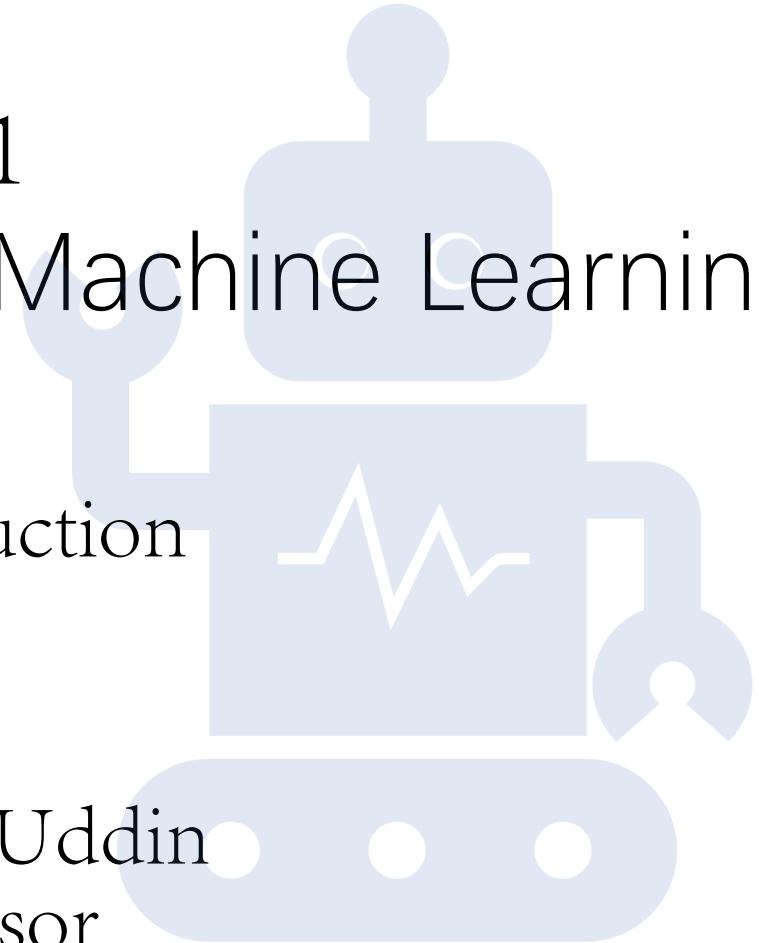
# Artificial Intelligence and Machine Learning



Lecture 1 - Introduction

Dr. A F M Shahab Uddin  
Assistant Professor

Department of Computer Science and Engineering  
Jashore University of Science and Technology





# Motivation/State-of-the-art

# What can we achieve with State-of-the-art AI / ML?

- **Teach a robot to interact with complex objects**





# What can we achieve with State-of-the-art AI / ML?

- Teach a robot to interact with complex objects
- **Teach a computer to play complex games**



# What can we achieve with State-of-the-art AI / ML?

- Teach a robot to interact with complex objects
- Teach a computer to play complex games
- **Teach a computer to describe images**

|                                                                                                                                     |                                                                                                                                         |                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <p>A young boy is playing basketball.</p>        | <p>Two dogs play in the grass.</p>                   | <p>A dog swims in the water.</p>        |
| <p>A group of people walking down a street.</p>  | <p>A group of women dressed in formal attire.</p>    | <p>Two children play in the water.</p>  |
| <p>A skier is skiing down a snowy hill.</p>    | <p>A little girl in a pink shirt is swinging.</p>  | <p>A dog jumps over a hurdle.</p>     |



# What can we achieve with State-of-the-art AI / ML?

- Teach a robot to interact with complex objects
- Teach a computer to play complex games
- Teach a computer to describe images
- **Robots that adapt**



# Autonomous vehicles



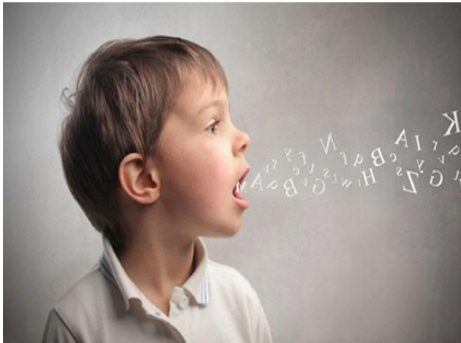
Aftenposten 2022-01-09, Arkivfoto: Siri Øverland Eriksen



New York Times, Nov 21, 2023

Speech ↔ text

Speech-to-text means converting spoken audio into written text using machine learning models. Modern AI models (like Whisper, wav2vec 2.0) can understand human speech with high accuracy.



und Polarisation des Lichtes in den sogenannten kolloidalen Metallösungen. Zieht man die Folgerungen aus der elektromagnetischen Lichttheorie auf das Verhalten der trüben Medien, so kommt man zu verschiedenen Resultaten, je nachdem die trübenden Teilchen Isolatoren oder Leiter der Elektrizität sind. Die bezüglichlichen Rechnungen sind durchgeführt worden von Lord Rayleigh<sup>1)</sup> für Isolatoren und von J. J. Thomson<sup>2)</sup> für Leiter der Elektrizität. Dabei machen beide die Annahme, daß die kleinen Teilchen Kugeln mit gegen die Lichtwellenlänge kleinem Durchmesser sind. Beide Autoren behandeln das Problem der Zerstreuung des Lichtes durch eine solche kleine Kugel, wenn diese von einer Welle natürlichen Lichtes getroffen wird. Während nun die Rechnung ergab, daß das von einer isolierenden Kugel in einer Ebene senkrecht zum einfallenden Strahl zerstreute Licht vollkommen linear polarisiert ist, und zwar in der durch die betrachtete Zerstreuungsrichtung



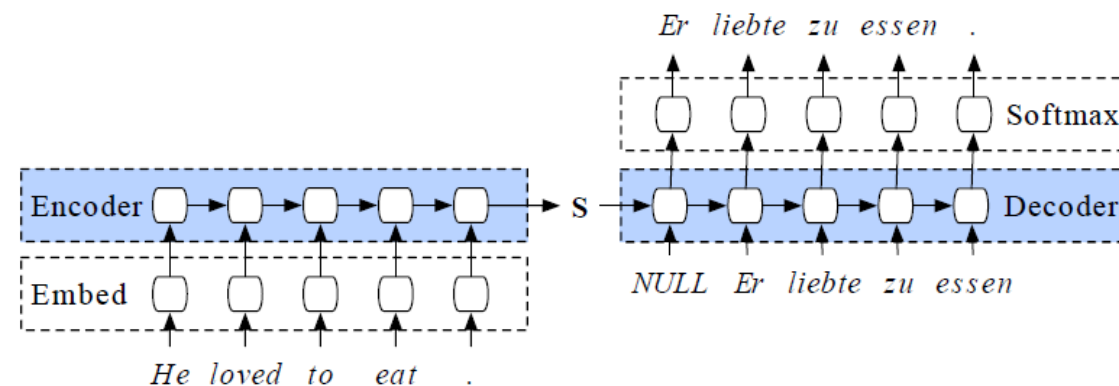


# Machine translation

9

10/31/2018

[https://smerity.com/media/images/articles/2018/gnmt\\_arch\\_1\\_enc\\_dec.svg](https://smerity.com/media/images/articles/2018/gnmt_arch_1_enc_dec.svg)



ltg/no-en-translation like 1 Running Logs

App Files Community

## Norwegian-English translation

Norwegian (Bokmål)

Tirsdag klokka 17:00 la Vy ut en beskjed til sine passasjerer på Østlandet. De anbefaler at du vurderer om du må reise med tog onsdag. Årsaken er at det kan bli veldig mye snø og vind på Østlandet. Det vil gi utfordringer for togtrafikken, ifølge Vy.

Submit

English

Tuesday at 17:00 Vy posted a message to its passengers in Eastern Norway. They recommend that you consider whether or not to travel by train on Wednesday. The reason is the fact it may be very much snow and wind in eastern Europe. This will bring challenges for rail traffic, according to Vy.

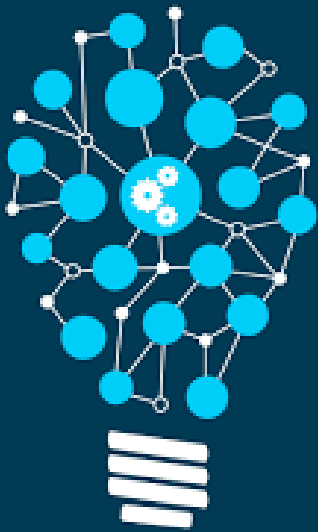


What will we learn?

Introduction to Artificial Intelligence  
Problem Solving and Search  
Knowledge Representation and Reasoning



MACHINE  
LEARNING



Introduction to Machine Learning  
Supervised Learning  
Unsupervised Learning  
Neural Networks and Deep Learning (Intro)  
Reinforcement Learning (Conceptual Overview)  
AI and ML Applications





# WHAT IS A.I.?

[Source: Wikipedia](#)

# The first use of the term "Artificial Intelligence"

- **Dartmouth summer research project on artificial intelligence: US, 1956**
- Organized by **John McCarthy, Marvin Minsky, Nathaniel Rochester, Claude Shannon**
  - *"The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."*
  - *"An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves."*
  - *"We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer. (2-month, 10 man)"*



Source: <https://www.historyofdatascience.com/>

# What is Intelligence?

(in humans:)

- No consensus definition, but
- "Mental abilities"
  - Aptitude – not acquired knowledge
  - More than one dimension
- Various lists of what to include
- Emphasis on differences



[Source](#)

- Are **animals** intelligent?
- Can **machines** be intelligent?
- Can a **chatbot** be intelligent?

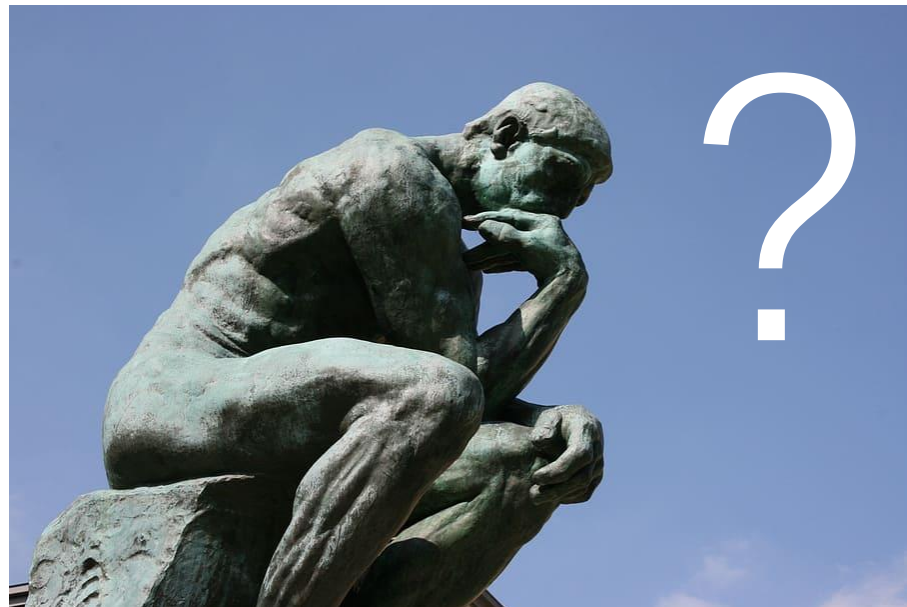


# What is Artificial Intelligence?

Definition?

"A machine is called intelligent if it can do tasks that normally require human intelligence when done by people."

But, one necessary condition for considering a task intelligent is that it is genuinely human – and cannot be carried out by an animal and even less a machine



## Definition of Artificial Intelligence along with its main types and examples

Artificial Intelligence (AI) is the technology that enables machines to perform tasks that normally require human intelligence such as learning, reasoning, problem-solving, and decision-making.

### Types of AI based on capability

#### 1. Narrow AI (Artificial Narrow Intelligence)

Works on one specific task and cannot go beyond its programming. Most AI systems today fall into this category. Examples: Siri, Google Assistant, Face Recognition, ChatGPT.

#### 2. General AI (Artificial General Intelligence)

An AI that can think, learn, understand, and solve problems like a human being. It can perform any intellectual task that a person can do. Examples: No real-world example yet; movie examples include JARVIS (Iron Man).

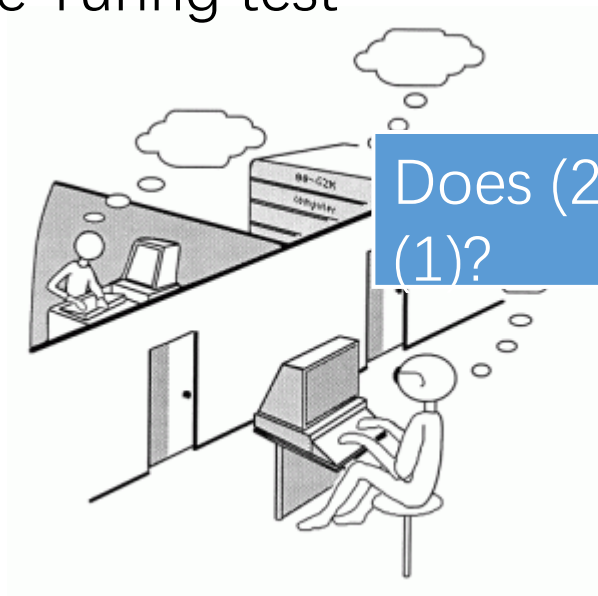
#### 3. Super AI (Artificial Super Intelligence)

AI that becomes more intelligent than humans in every field, including creativity, emotions, and decision-making. Fully theoretical and not yet developed. Examples: No real examples; fictional ones include Ultron and Terminator's Skynet.

# Goals of AI

## 1. Theoretical:

- Build models of how humans think and act
- Understand ourselves
- The Turing test



## 2. Practical:

- Replace human activities that demand intelligence
- Assist us by doing boring jobs





# Main approaches to AI

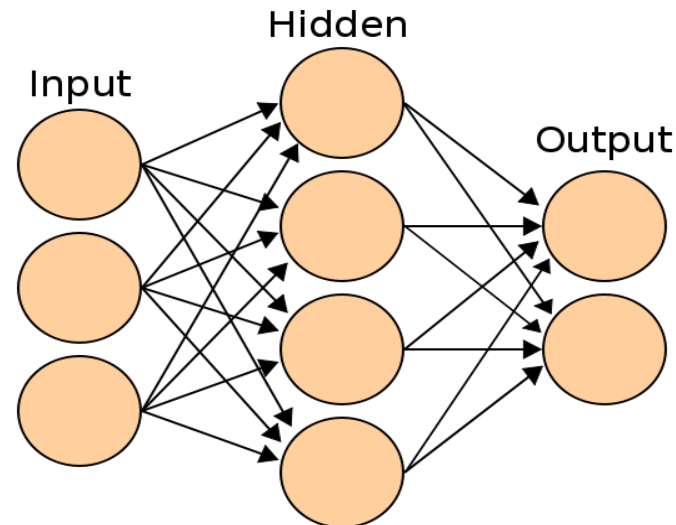
## Rule-based

- Logic, deduction
- Explicit coding of knowledge as formulas or rules
- Expert systems
- Dominated AI books until the end of the XX century

$\forall x(\text{human}(x) \rightarrow \exists y(\text{mother\_of}(y,x)))$

## Machine learning

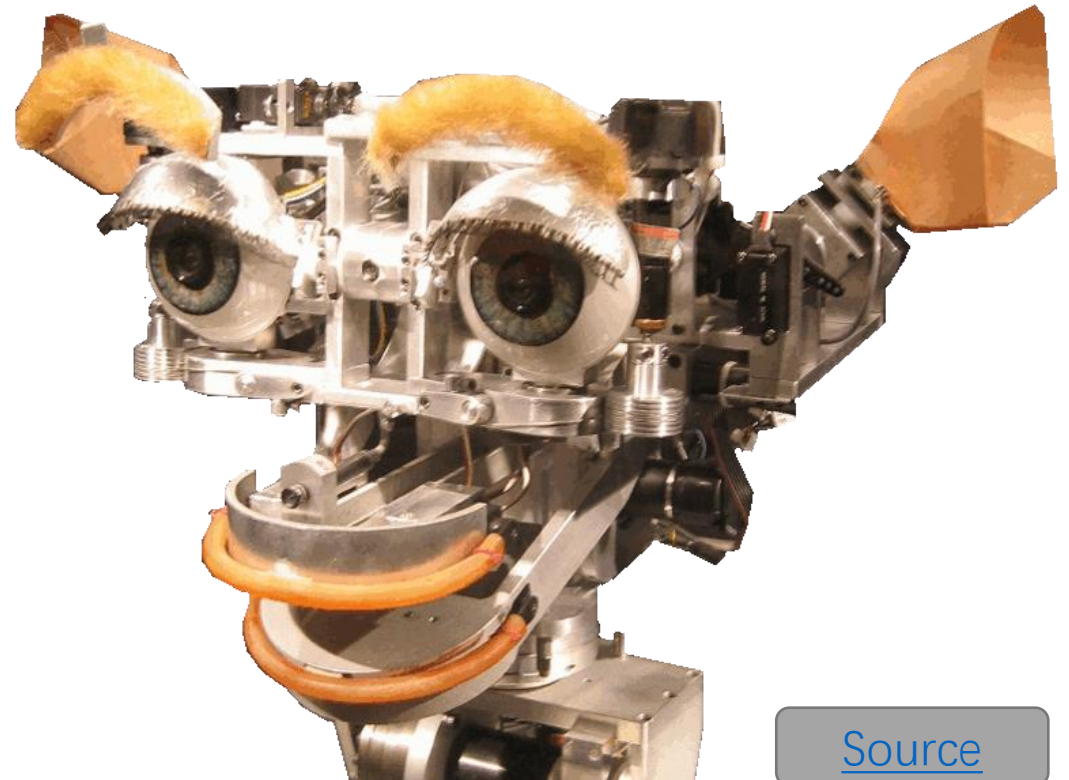
- Induction rather than deduction
- Adapt to the environment
- Now dominant
- Main focus in this course



[Source](#)

# Traditional areas of AI

- Problem solving
  - Search
  - Game playing
- Knowledge and Reasoning
  - Logic
  - Theorem proving
  - Knowledge representation
- Planning
- Learning
- Natural language understanding
- Natural language generation
- Perception
- Motion and manipulation

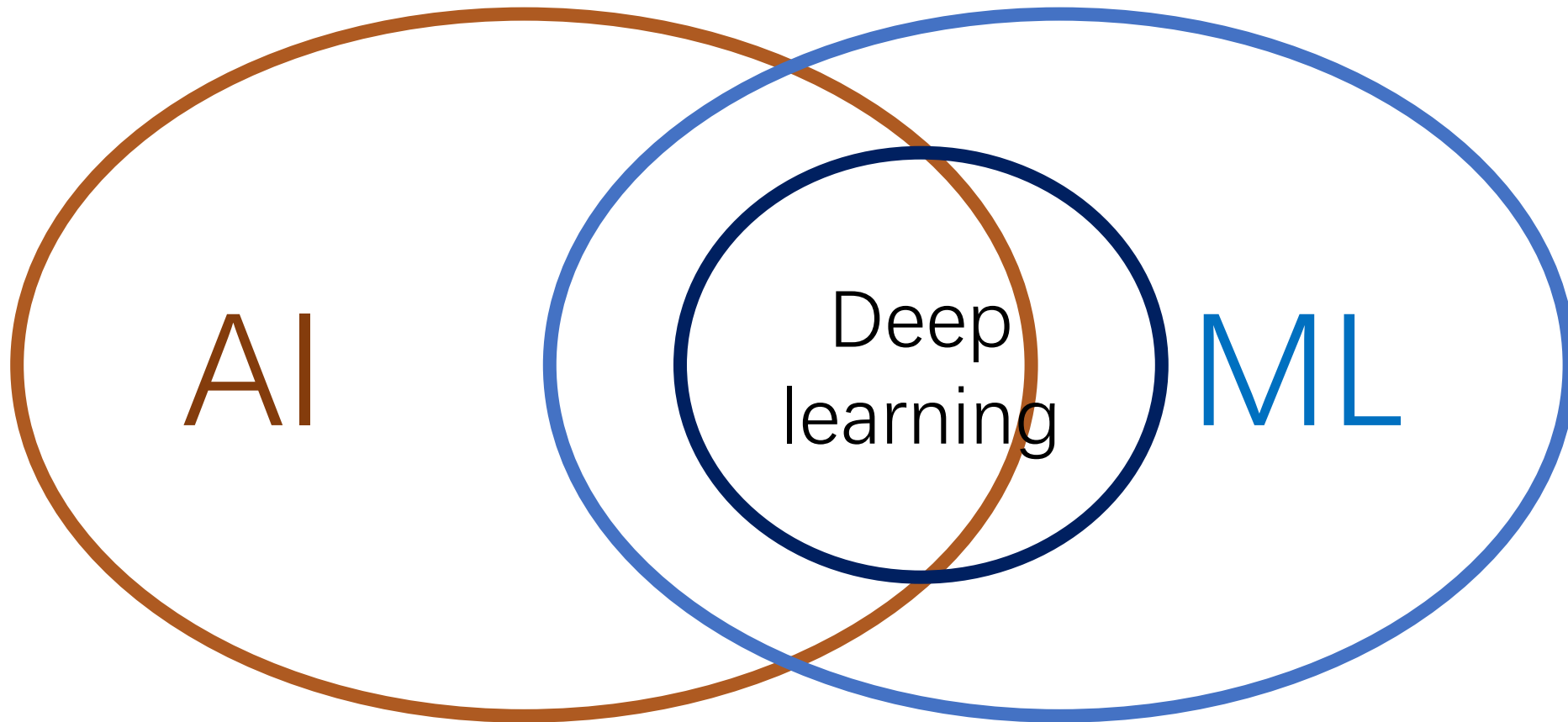


[Source](#)

## What AI can do in today's world?

Artificial Intelligence (AI) plays a major role in today's world by performing tasks that usually require human intelligence. It can recognize speech, understand commands, and convert voice into text, which is used in assistants like Siri and Google Assistant. AI can also analyze images and videos, enabling face recognition, medical image diagnosis, and object detection. In business, AI helps by analyzing large amounts of data to make predictions such as sales forecasting, fraud detection, and customer behavior analysis. AI-powered systems are used to automate tasks, including chatbots, robots in industries, and self-driving vehicles. AI also improves daily life through personalized recommendations in platforms like YouTube, Netflix, and Amazon. Overall, AI enhances efficiency, accuracy, decision-making, and automation across healthcare, education, banking, transportation, and entertainment.

# AI, Machine learning, Deep learning





# Artificial Intelligence

Is the field of study

## Machine Learning

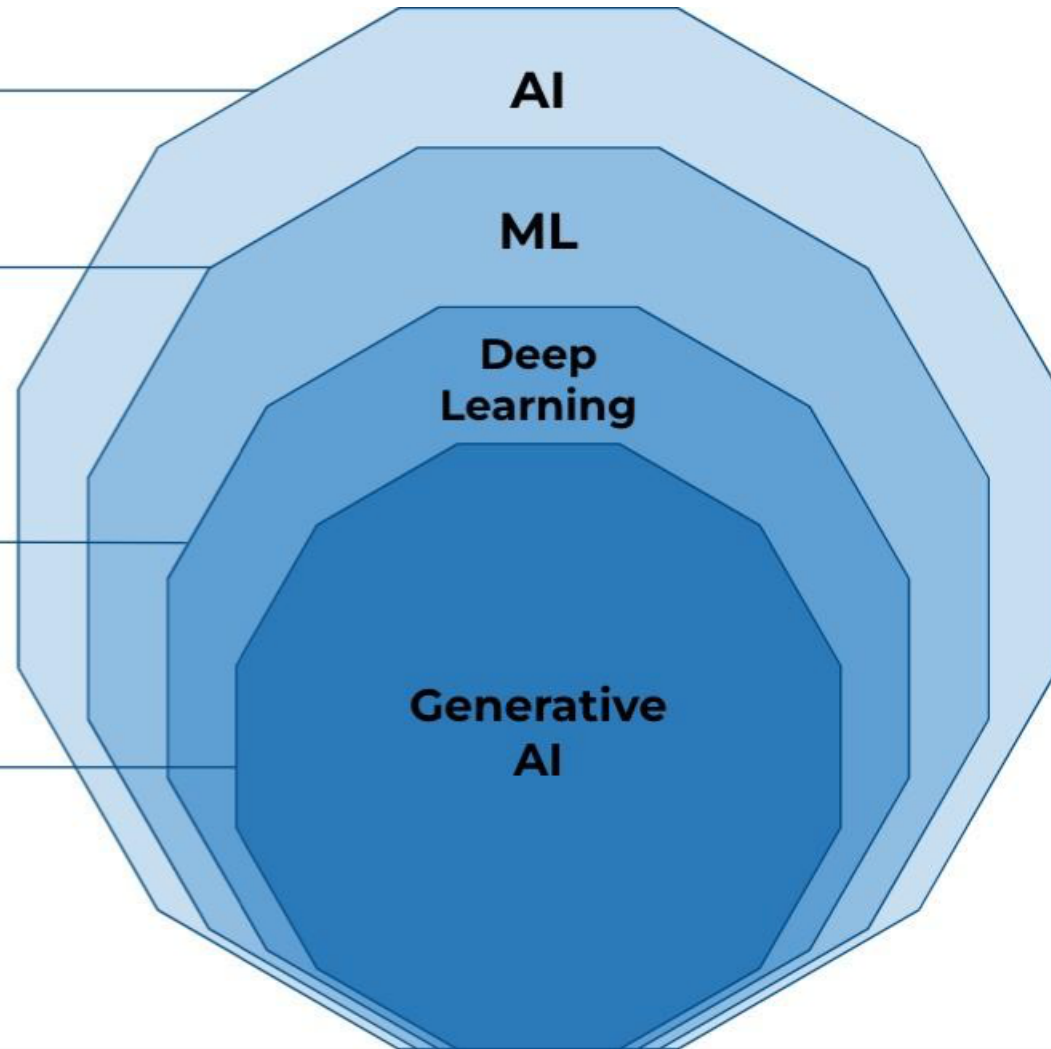
Is a branch of AI that focus on the creation of intelligent machines that learn from data.  
Another very well know branch inside AI is **Optimization**.

## Deep Learning

Is a subset of Machine Learning methods, based on **Artificial Neural Networks**.  
Examples: CNNs, RNNs

## Generative AI

A type of ANNs that generate data that is similar to the data it was trained on.  
Examples: GANs, LLMs



Human vs Machine – Who is smartest? Some tasks machines are better at

- **Math/logic**

$$\nabla \dot{\xi} \cdot \frac{\partial^\varepsilon \chi}{\partial \mathbf{p}^\varepsilon} + \nabla \mathfrak{J} \wedge \frac{\partial^\gamma \psi}{\partial \mathbf{q}^\gamma} = 0$$

$$\oint_{\text{全てのスペース}} (E + H \wedge T) \int_{-\infty}^{+\infty} \frac{\partial^2 \mathcal{R}}{\partial \phi \partial z} d\Omega d\tau = \frac{\Gamma(\mathcal{N}) \mathfrak{J}(\Omega, \tau)}{(2\pi)^{\mathcal{N}} \mathbb{K}}$$

$$d\mathbf{F} = \frac{\langle \Phi | \mathfrak{J} | \Psi \rangle}{(2\pi)^{\mathcal{N}} c^2} \left[ \gamma d\mathbf{\Sigma} + \mathbf{b} \frac{\partial \mathfrak{Z}}{\partial z} \wedge d\dot{\xi} \right]$$

# Some tasks machines are better at

- Math/logic
- **Board games (chess, go, etc)**



# AI learns the art of Diplomacy

Meta's algorithm tackles both language and strategy in a classic board game that involves negotiation

22 NOV 2022 • 10:00 AM • BY [MATTHEW HUTSON](#)





# Some tasks machines are better at

- Maths/logic
- Board games (chess, go, etc).
- **Some complex computer games**
  - Starcraft II and Dota 2 have been the focus of a lot of research in Reinforcement Learning
  - **Minecraft** is also a popular testbed due to its open-ended nature.



[Dette bildet](#) av Ukjent forfatter er lisensiert under [CC BY-SA](#)





## ARTIFICIAL INTELLIGENCE

# A bot that watched 70,000 hours of Minecraft could unlock AI's next big thing

Online videos are a vast and untapped source of training data—and OpenAI says it has a new way to use it.

By Will Douglas Heaven

November 25, 2022

**Log**

280 actions  
14 seconds  
99% 10 min.

**Planks**

870 actions  
44 seconds  
99% 10 min.

**Crafting Table**

970 actions  
49 seconds  
99% 10 min.

**Wooden Pickaxe**

1390 actions  
1.2 minutes  
99% 10 min.

**Cobblestone**

2050 actions  
1.7 minutes  
99% 10 min.

**Stone Pickaxe**

2790 actions  
2.3 minutes  
98% 10 min.

**Iron Ore**

6540 actions  
5.4 minutes  
84% 10 min.

**Furnace**

7320 actions  
6.1 minutes  
78% 10 min.

**Iron Ingot**

10870 act.  
9 minutes  
60% 10 min.

**Iron Pickaxe**

11161 act.  
9.3 minutes  
57% 10 min.

**Diamond**

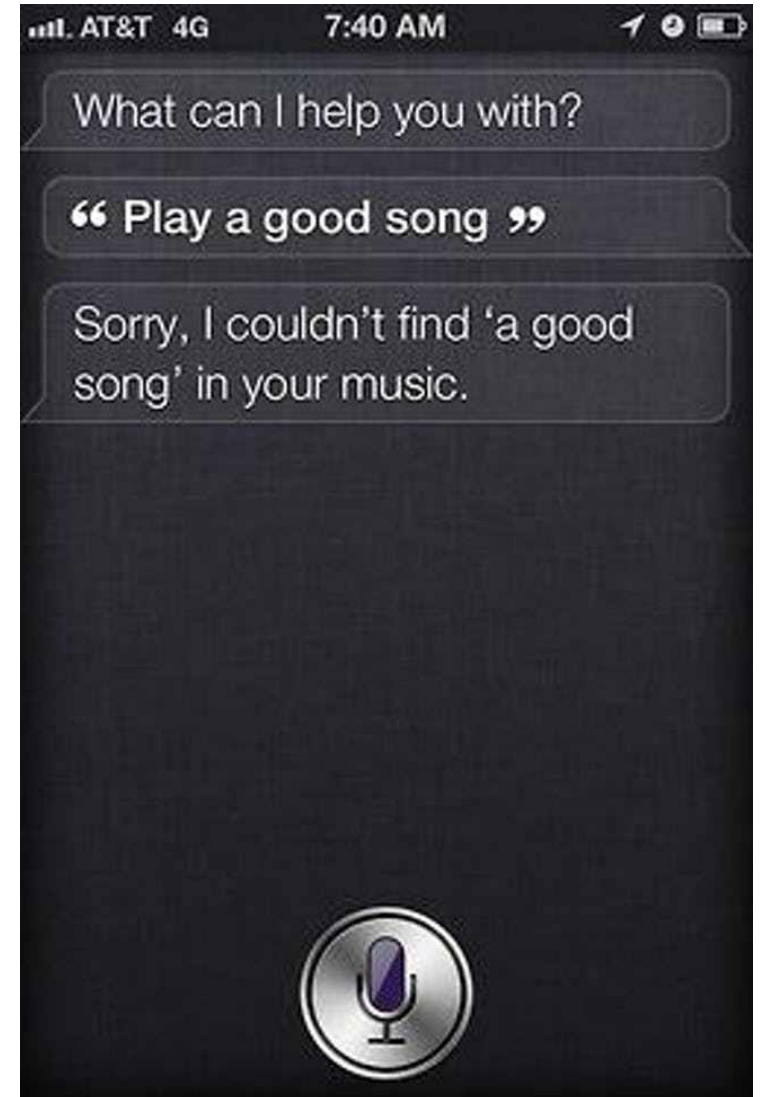
23975 act.  
20 minutes  
15% 10 min.

**Diamond Pickaxe**

24000+ act.  
20+ minutes  
12% 10 min.

# Some tasks humans are better at

- **Communicating with other humans**





# Large language models: Towards human-level language understanding?



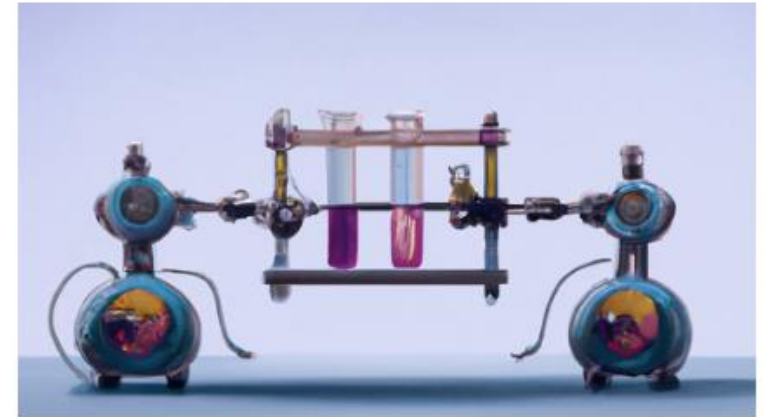
**BIG** THINK

SUBSCRIBE

THE FUTURE — DECEMBER 8, 2022

## We gave ChatGPT a college-level microbiology quiz. It blew the quiz away.

ChatGPT's capabilities are astonishing.



Credit: Annelisa Leinbach, DALL-E AI

LLM (Large Language Model) is a type of advanced AI system that is trained on a massive amount of text data so it can understand, generate, and use human language in a meaningful way.

<https://bigthink.com/the-future/chatgpt-microbiology-quiz-aced/>

# But there are still important limitations



What weighs more: Two pounds of bricks or one pound of feathers?

Chat with Bing Chat January 2023

What weighs more: Two pounds of bricks or one pound of feathers?

✓ Søker etter: two pounds of bricks or one pound of feathers which weighs more

✓ Genererer svar for deg ...



Dine personlige data og firmadata er beskyttet i denne chatten

Both two pounds of bricks and one pound of feathers weigh the same, which is 2 pounds<sup>1</sup>. However, the volume of two pounds of bricks is much smaller than one pound of feathers. This is because bricks have a higher density than feathers<sup>2</sup> <sup>3</sup>.

I hope that helps!

Finn ut mer

1

 [wyzant.com](https://www.wyzant.com)

2

 [short-fact.com](https://short-fact.com)

3

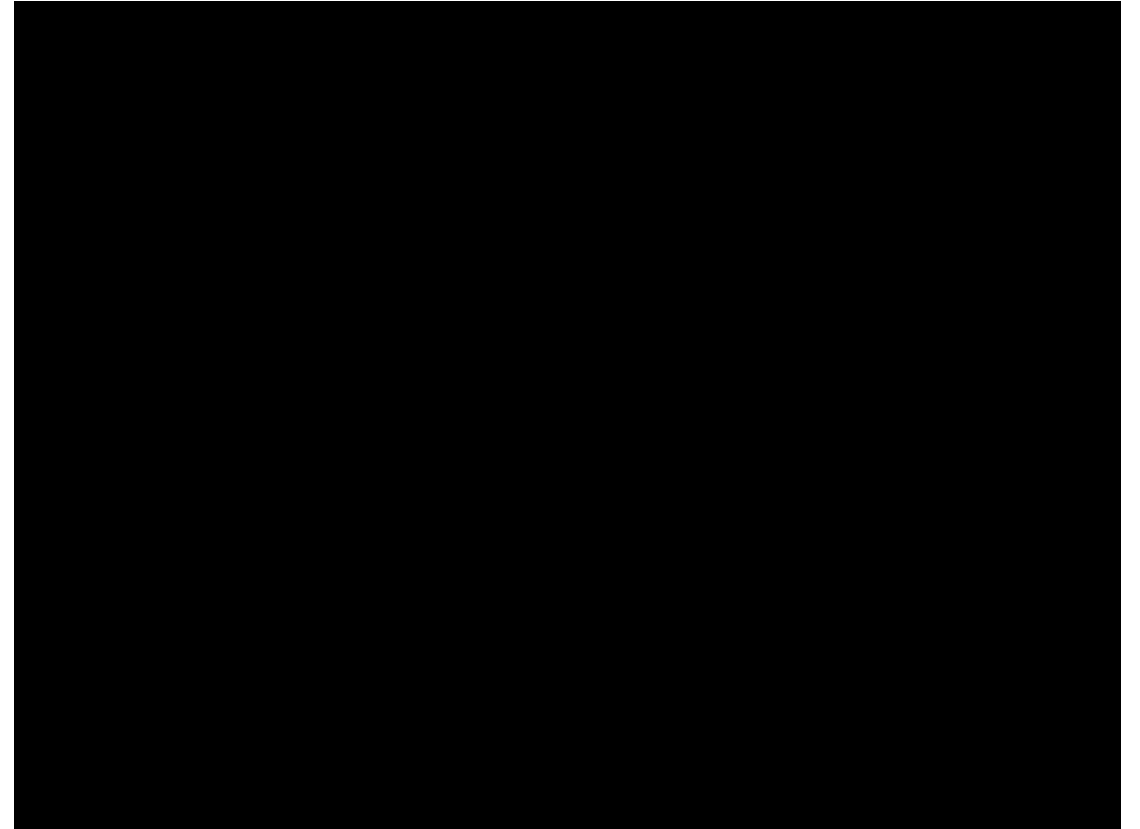
 [wyzant.com](https://www.wyzant.com)



5 av 30 svar ●

# Some tasks humans are better at

- Communicating with other humans
- **Moving around**





# Some tasks humans are better at

- Communicating with other humans
- Moving around
- **Understanding from vision**
  - Machines are getting close to human performance

**GPT-4 visual input:**

What is funny about this image? Describe it panel by panel.





Source: <https://www.reddit.com/r/hmmm/comments/ubab5v/hmmm/>

GPT-4 The image shows a package for a "Lightning Cable" adapter with three panels.

Panel 1: A smartphone with a VGA connector (a large, blue, 15-pin connector typically used for computer monitors) plugged into its charging port.

Panel 2: The package for the "Lightning Cable" adapter with a picture of a VGA connector on it.

Panel 3: A close-up of the VGA connector with a small Lightning connector (used for charging iPhones and other Apple devices) at the end.

The humor in this image comes from the absurdity of plugging a large, outdated VGA connector into a small, modern smartphone charging port.



# The state of Computer Vision and AI: we are really, really far away.

Oct 22, 2012





Can you explain to me why someone might find this image funny?







- ✓ Analyserer bildet: Ansikter kan være uskarpe for å beskytte personvernet
- ✓ Søker etter: why someone might find an image funny
- ✓ Genererer svar for deg ...

 Dine personlige data og firmadata er beskyttet i denne chatten

I'm sorry, but I cannot infer the context of the image from the description you provided. However, humor is a complex and subjective phenomenon that can be influenced by various factors such as culture, personality, and context <sup>1</sup>. It is possible that the image contains elements that are unexpected, absurd, or ironic, which can trigger a humorous response in some individuals.

If you could provide me with more information about the image or the context in which it was shared, I might be able to provide a more informed opinion. Alternatively, you could try sharing the image with your friends or colleagues to see if they find it funny. Sometimes, humor can be contagious and shared laughter can be a great way to bond with others <sup>1</sup>.

Finn ut mer  1  [psychologytoday.com](https://psychologytoday.com) 2  [boredpanda.com](https://boredpanda.com) 3  [freethink.com](https://freethink.com)



1 av 30 svar 

# Some tasks humans are better at

- Communicating with other humans
- Moving around
- Visual tasks
- **Creative tasks**
  - **Machines are improving**

# A painting of a fox sitting in a field at sunrise in the style of claudes monet

Dall-E 1 (2021)



Dall-E 2 (2022)



Dall-E 3 (2023)

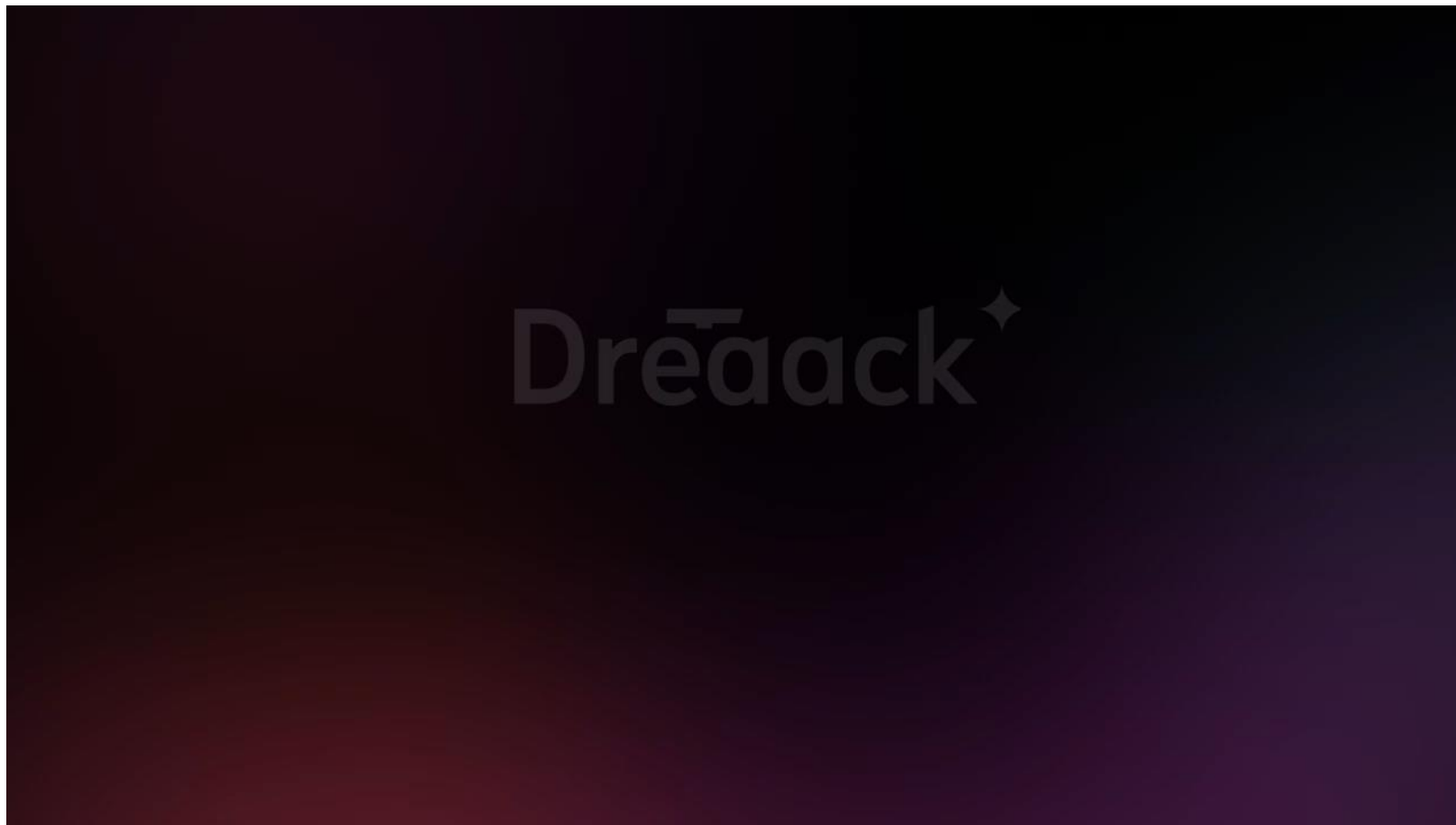








AI for music is also coming



# Moravec's Paradox: What is easy for humans is hard for machines

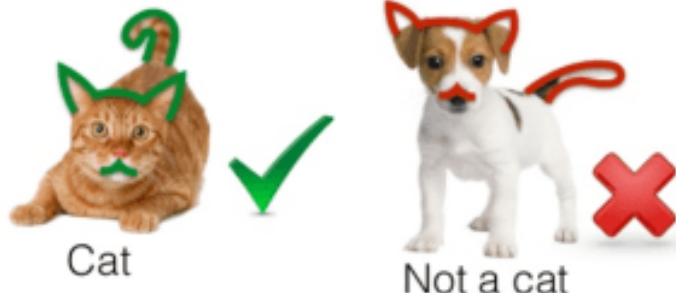
- Robotics researcher Hans Moravec (1988):  
«it is comparatively **easy** to make computers exhibit adult level performance on **intelligence tests or playing checkers**, and **difficult** or impossible to give them the skills of a one-year-old when it comes to **perception and mobility**»



Wikipedia

# Moravec's Paradox

Humans

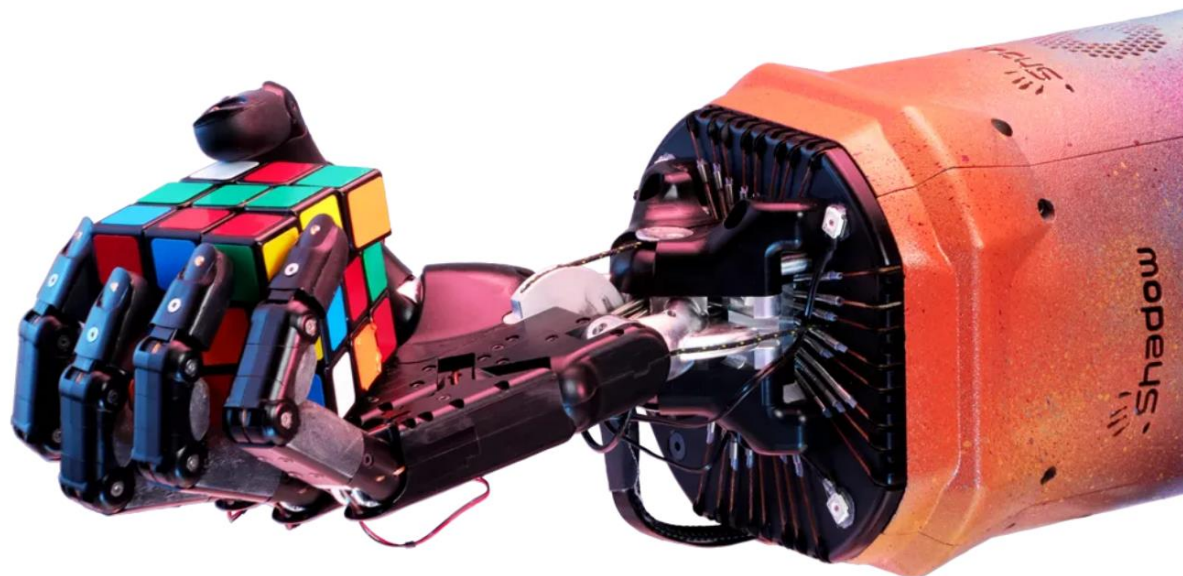


Machines

$$\nabla_{\xi} \cdot \frac{\partial^{\varepsilon} \chi}{\partial \mathbf{p}^{\varepsilon}} + \nabla_{\mathbf{j}} \wedge \frac{\partial^{\gamma} \psi}{\partial \mathbf{q}^{\gamma}} = 0$$

$$\oint_{\text{全てのスペース}} (E + H \wedge T) \int_{-\infty}^{+\infty} \frac{\partial^2 \mathcal{H}}{\partial \phi \partial z} d\Omega d\tau = \frac{\Gamma(\mathcal{H}) \mathcal{J}(\Omega, \tau)}{(2\pi)^{\prime\prime} \mathcal{K}}$$

$$\mathcal{H} \mathbf{F} = \frac{\langle \Phi | \mathcal{J} | \Psi \rangle}{(2\pi)^{\prime\prime} c^2} \left[ \gamma d\mathbf{\Sigma} + \mathbf{b} \frac{\partial \xi}{\partial z} \wedge d\xi \right]$$



# Next week: Search/Optimization

- Optimization: A principle behind most of the other topics in this course
- We'll talk about what optimization problems are and begin to think about ways to solve them
- You will also test this yourselves in Mandatory Exercise 1

